6 **(a)** Scattering forces tend to push the particle in the direction of the beam. The gradient forces must be greater in magnitude so that if the particle is displaced from the focal point along the direction of the beam, there will be a net force to attract towards the focal point.

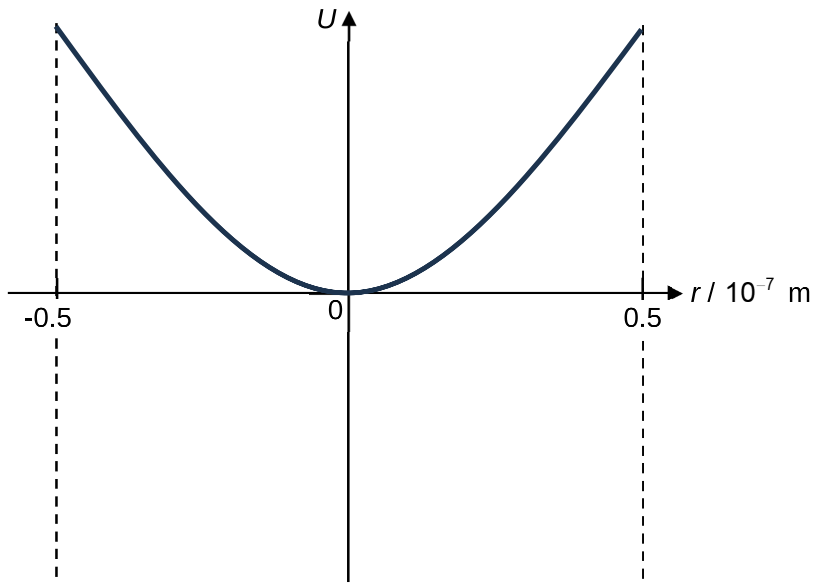
(b) **(i)** 0.70×10^{-7} m or 7.0×10^{-8} m

(ii) $F = kx$
$3.5 \times 10^{-12} = k(0.70 \times 10^{-7})$
$k = 5.0 \times 10^{-5}$ N m^{-1}

(iii) 1. Straight line / Constant gradient passing through the origin indicates force is proportional to displacement from equilibrium position. Negative gradient indicates force is directed towards the equilibrium position

2. $a = - \omega^2 x$
$(3.5 \times 10^{-12}) / (1.50 \times 10^{-14}) = - (2\pi f)^2(0.70 \times 10^{-7})$
$f = 9189 = 9190$ Hz

(c)



Quadratic curve (U or N shaped curve)
Positive and intersect at origin

(d) Incident photon has no momentum along the radial direction initially. Refracted photon through the glass bead gains momentum to the left. By Newton's second law, refracted photon experiences a force towards the left. By Newton's third law, the glass bead experiences a force towards the right

OR

Incident photon has no momentum along the radial direction initially. Refracted photon through the glass bead gains momentum to the left. By conservation of momentum, glass bead gains momentum to the right. By Newton's second law, the glass bead experiences a force towards the right

(e) **(i)** $p = h / \lambda$
$= (6.63 \times 10^{-34}) / (960 \times 10^{-9})$
$= 6.91 \times 10^{-28}$ kg m s^{-1}

(ii) impulse $= p_f - p_i$
$= (6.91 \times 10^{-28} \times \cos 10°) - (6.91 \times 10^{-28} \times \cos 32°)$
$= 9.44 \times 10^{-29}$ kg m s^{-1}

(f) Some lights / photons are reflected off the glass bead resulting in the scattering force in the positive axial direction. This force due to reflection will in a larger net restoring force in the positive axial direction (when glass bead is displaced negatively)